

DANG TRAN NAM - Software Developer Intern

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Summary

A detail-oriented Software/DevOps Engineer with a strong interest in backend/cloud architecture and performance optimization. I strive to build reliable, high-efficiency systems by understanding the fundamental interactions between software and hardware. Committed to writing clean, maintainable code and continuously optimizing system latency through data-driven approaches.

Experience

An Binh Hospital Jan 2026 - Present
Software Engineer Intern

IT ASSET MANAGEMENT (ITAM) – DISTRIBUTED TELEMETRY SYSTEM

- **Distributed Agent Architecture:** Developed a cross-platform agent (Go) that interfaces directly with OS APIs (Windows/Linux) to harvest real-time CPU/RAM/Disk metrics.
- **Resilience & Fault Tolerance:** Engineered a **Circuit Breaker** pattern with exponential backoff and "Jitter" logic to prevent Thundering Herd effects when thousands of agents reconnect simultaneously.
- **Resource Efficiency:** Implemented strictly typed log rotation and data pruning strategies, ensuring the agent operates within a minimal memory footprint (<15MB RAM) on edge devices.
- **Security:** Secured Agent-Server communication via Bearer Token authentication and SHA-256 HMAC signature verification for critical commands (Kill Switch).

Education

Saigon University (SGU) 2021-09-01 - 2026-06-01
Software Engineering Engineer

Projects

DONALD VIBE – HIGH-CONCURRENCY INVENTORY ENGINE

Solo Architect A specialized transactional engine designed to handle "thundering herd" traffic spikes during limited drop.

- **Cloud Infrastructure & Emulation:** Automated the provisioning of AWS services (S3, SQS, Secrets Manager) for local development and CI pipelines using **LocalStack** and **Bash scripting**, ensuring environment parity and seamless cloud-native integrations without incurring actual AWS costs.
- **Containerization & Orchestration:** Engineered lightweight (~15MB), statically linked, multi-stage **Docker** builds on Alpine Linux. Orchestrated multi-tier deployment topologies (Load Balancer, App Replicas, Cloud Emulator) across Dev, Test, and CI environments using **Docker Compose**.
- **Traffic Routing & OS-Level Tuning:** Configured **Nginx** as a reverse proxy and load balancer to manage "thundering herd" traffic spikes across stateless Go instances. Optimized system-level I/O by executing C-level database tuning (Write-Ahead Logging, memory-mapped files), enabling the system to sustain **10,000+ writes/sec** with p95 latency under **0.1ms**.

Skills

Hard Skills

Linux (Arch/Ubuntu), Golang, Rust, JavaScript, Docker, C/C++, Python, AWS, Google Cloud Platform

Certifications

TOEIC (760 R/L) **Software Engineer Intern Certificate - HackerRank**
AWS Cloud (In Progress)